ClipSAT : IB Diploma Mathematics: Analysis & Approaches / Applications [ANSWER KEY]

Unit 1.01: Topic 1: Number & Algebra Sequences, Logs & Binomial

Q1. Consider the function fx = 2x + 5 . What is the value of f3 ?

[x] A: 11
[ ] B: 8
[ ] C: 10
[ ] D: 16

Solution: To find f3 , substitute x=3 into the function fx = 2x + 5 . f3 = 23 + 5 = 6 + 5 = 11 . [M1] [A1]

Q2. The graph of the quadratic function fx = x^2 - 6x + 5 is shown below. Find the value of the y-intercept.

[x] A: 5
[ ] B: -5
[ ] C: 0
[ ] D: 1

Solution: The y-intercept of a function occurs where the graph crosses the y-axis, which is when x = 0 . For the function fx = x^2 - 6x + 5 , when x = 0 , we have f0 = 0^2 - 60 + 5 = 0 - 0 + 5 = 5 . Therefore, the y-intercept is 5. [M1] [A1]

Q3. Consider the function fx = x^2 - 4x + 3 and the line gx = x - 1 . The graph of these functions is shown below. Find the coordinates of the point where the line intersects the parabola.

[x] A: 1, 0
[ ] B: 3, 2
[ ] C: 2, 1
[ ] D: 4, 3

Solution: To find the intersection points, we set the two functions equal to each other: fx = gx x^2 - 4x + 3 = x - 1 x^2 - 5x + 4 = 0 Factoring the quadratic equation: x-1x-4 = 0 This gives us two possible x-values for the intersection points: x=1 or x=4 . nUsing gx = x-1 to find the corresponding y-values: nFor x=1 : y = 1 - 1 = 0 . The intersection point is 1, 0 . nFor x=4 : y = 4 - 1 = 3 . The intersection point is 4, 3 . nLooking at the graph, one of the intersection points is 1, 0 . [M1] [A1]

Q4. e^0=

[ ] A: 0
[x] B: 1
[ ] C: e
[ ] D: âˆž

Solution: Any nonzero number to power 0 is 1. [M1] [A1]

Q5. The graph of the quadratic function fx = ax^2 + bx + c is shown below. It passes through the points 0, -3 , 1, 0 , and -1, -4 . Find the value of c .

[x] A: -3
[ ] B: 3
[ ] C: -1
[ ] D: 1

Solution: The general form of a quadratic function is fx = ax^2 + bx + c . The graph of the function passes through the point 0, -3 . Substituting these coordinates into the function, we get: f0 = a0^2 + b0 + c . Since f0 = -3 , we have -3 = 0 + 0 + c , which means c = -3 . [M1] [A1]

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Q6. Consider the function fx = 2x - 3 . Determine the value of f3 .

[x] A: 3
[ ] B: 6
[ ] C: 9
[ ] D: 0

Solution: To find the value of f3 , substitute x=3 into the function fx = 2x - 3 .
f3 = 23 - 3 f3 = 6 - 3 f3 = 3 [M1] [A1]

Q7. Solve for x : 3^x=9 .

[ ] A: 3
[ ] B: 4
[x] C: 2
[ ] D: 1

Solution: Working through this algebra step: 3^x=9=3^{2} Rightarrow x=2 .
[M1] [A1]

Q8. An arithmetic sequence has first term u_1=7 and common difference d=2 . Find u_{15} .

[ ] A: 37
[x] B: 35
[ ] C: 33
[ ] D: -21

Solution: u_n=u_1+n-1d Rightarrow u_{15}=7+15-12=35 . [M1] [A1]